Mark	

Team name:	*A10*
Homework number:	*HOMEWORK 02*
Due date:	02/10/2026

Contribution	NO	Partial	Full
Carbone Alessandro			*X*
Casareale Alberto			*X*
Colombi Federico			*X*
Lorusso Luca			*X*
Martorano Gabriele			*X*
Notes: Everyone tried by himself and successfully completed the task			

Project name	Toggle the green LED on each finger snap		
Not done	Partially done (major problems)	Partially done (minor problems)	Completed
			X

Explanation:

Part 1: Hardware Routing and Configuration

To achieve the goal of toggling the green LED upon detecting a sound, we first consulted the "hands-on schematic" to identify the correct hardware routing. The schematic confirmed that the microphone's output signal (SND_IN) is physically connected to the PA8 pin of the microcontroller.

In the STM32CubeMX software, we configured the PA8 pin as an external interrupt (GPIO_EXTI8) and set its trigger detection to respond to both rising and falling edges. To ensure the microcontroller processes these signals, we enabled the corresponding interrupt by checking the enable box in the NVIC settings. Additionally, regarding the blue user push-button mapped to pin PC13, we physically set it to a reset state on the board to disable it, eliminating the need to handle its logic in the code.

Part 2: Code Generation and Verification

After generating the project code, we inspected the stm32f4xx_it.c file to verify that the interrupt request handler for pin 8 was automatically and correctly set up by CubeMX.

Part 3: Implementation of Functional Logic

Finally, we implemented the functional logic inside the external interrupt callback function. This function is triggered automatically by the hardware whenever the microphone detects a sound. Because the microphone lacks an analog hardware debouncer, a single sound like a finger snap produces a noisy signal that causes multiple rapid interrupts.

To handle this, the code uses a software debounce mechanism based on the system timer. It records the exact time the first peak of the sound is processed and strictly ignores any subsequent interrupt signals

that occur within 200 milliseconds. When a valid, distinct sound passes this time filter, the code simply toggles the state of the green LED on pin PA5, turning it on if it was off, and off if it was on. This approach ensures a clean and reliable state change for every individual snap, without wasting CPU resources on continuous polling.

Professor comments: